

Looking closely

The invention of the microscope revealed a whole new world.

For the first time, scientists could observe objects too small to be seen with the naked eye. Researchers began to understand the building blocks of life as they studied the structure of cells and discovered the existence of microorganisms. Today, microscopes can even identify individual atoms.



19th-century poster celebrates Janssen and his invention of the microscope

The first microscope

By the 1590s, Dutch spectacle-makers were making microscopes by fixing two lenses in a tube. They found that light, refracted by the two lenses, made objects larger than a single lens did on its own. One of these spectacle-makers, Zacharias Janssen, may have made the first microscope.

Key events

1590

Zacharias Janssen is usually credited with the invention of the first compound microscope.

1661

Italian biologist Marcello Malpighi saw red blood cells, which he called particles, through a microscope.

1665

Robert Hooke published *Micrographia*. It contained illustrations of the tiny objects he had observed through a microscope.

1674

Antoni van Leeuwenhoek designed a single-lens microscope that was able to magnify objects up to 270 times.

1860

French chemist Louis Pasteur used a microscope to carry out research into disease-carrying microbes.

Scientific bestseller

Micrographia, published by Robert Hooke in 1665, introduced the public to the world of the microscope. It included wonderfully detailed drawings of the objects observed through his microscope, including fleas, hairs, and even a fly's eye. The book became an instant bestseller.

Hooke's illustration of a flea



Water-filled sphere focuses light from oil lamp.

Wooden barrel

Discovering cells

Robert Hooke, a noted English scientist, was the curator of experiments at the Royal Society in London, England. He designed a compound microscope that used a water-filled sphere to focus light from an oil lamp onto the specimen. Hooke noticed the spaces between long empty cell walls in a piece of cork he was examining and first coined the word "cell" to describe them.

Lens concentrates light onto specimen.

Specimen mounted on pin

Hooke's compound microscope

